

Coral Bleaching Game User Guide

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Overview



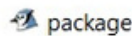
Welcome to the Coral Bleach Game, a simulation game that aims to teach users how different choices that businesses make can impact coral reefs, aquatic ecosystems, and people!

Using the user guide, users can better understand the features that this game has to offer, enriching a fun experience!

System Requirements

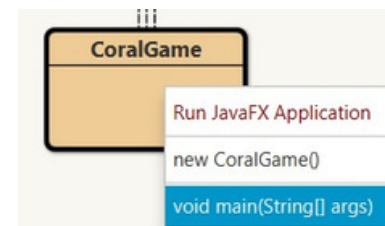
- BlueJ
- A computer running Windows, or macOS

Installation

- Download BlueJ following instructions at <https://www.bluej.org/>
- Download and unzip the Coral Game file by right clicking on the zipped file and pressing “extract all” 
- Once the folder is extracted, open it and find the option that says “Package”. Press to open the project 

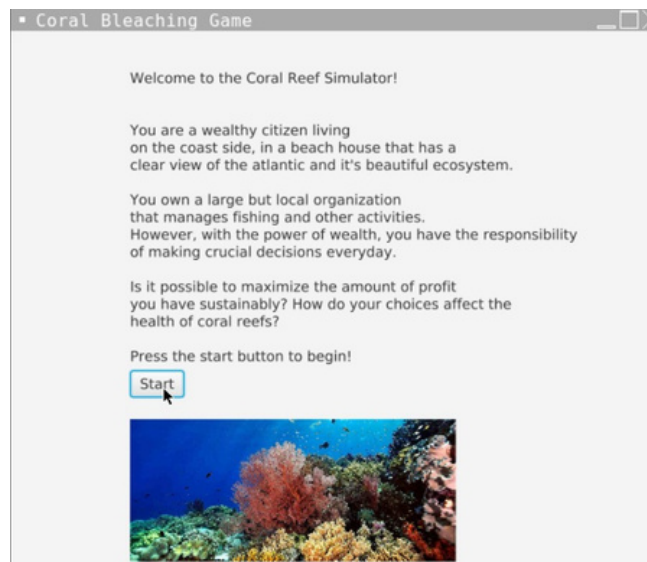
Getting Started

- To begin, right click on the CoralGame class and press the third option down, which says “void main(String args[])”.
- This option will open up a window that says Method call. Do not type. Press the enter key to open the game.
- The game window will open in a few seconds. If you cannot find the game, open all your windows to see if the game is under another window.



GamePlay

The Coral Bleaching Game begins with a start screen, where it will give the user a blurb of the scenario that they're in. To start the game, press the “Start” button.



Scenarios

You're business is booming! You find that you may need to build a headquarters somewhere but the only land available near you is a forest.

Build Headquarters

Don't Build Headquarters

There is a drought in the area. Water levels are falling and hurting the coral reefs. Your business is losing money.

Work on rebuilding your company.

Donate money to help with the ocean

Throughout the game, there will be several scenarios that you face. You will be offered two buttons for each scenario. One choice will make you lose money and lower temperature, increasing coral health, while the other will make you gain money and increase temperature, lowering coral health.

The game also has a factor of **Natural Disasters**! Be careful, the choice you make for these is going impact the status bar variables much more heavily than the other scenarios! (The chance of getting a natural disaster scenario is 6.6%.)

Status Bars

There are 5 variables that the game tracks:

- **Money**: how much money the user has (Initial = \$1000)
- **Coral Health**: the health of the coral depends on the temperature status and translates to the user the health of the corals (Initial = Healthy)
- **Day**: number of days (how many choices you've made) (Initial = 0)
- **Temperature**: the temperature of the water in the ocean in Fahrenheit (Initial = 78)
- **Coral Color**: the color that the corals would be depending on the temperature (because corals are very sensitive to change in temperature!) (Initial = Light Red)

Game Over

There are two ways for the user to lose the game.

- **Money**: \$0
- **Temperature**: 104.0°F (or Coral Health: Dead)

Once one of these statuses are achieved, you will have to press one of the buttons one more time to reach the game over screen. It will display all of your statuses. To start over, press the "Start Over" button. This will reset all your statuses and display scenarios and options again.

Money: \$1400.0

Coral Health: Healthy

Temperature: 82.0°F

Day: 8



FAQ

Q: Can I play the Coral Bleaching Game without an internet connection?

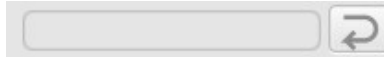
A: Yes!

Q: What should I do if the game crashes?

A: Close out of BlueJay and open the game file again.

Q: BlueJay is displaying error when following the instructions to open the game window.

A: Close out of Bluejay and open the game file again. To prevent this error from occurring, make sure you are closing the game window before calling it to open again. Once closed, press on the arrow at the bottom right side of the BlueJay screen.



This should initialize the virtual machine and allow you to call for the game window again.

Q: At the same temperature, the coral is a different color than it was before. Why?

A: Corals are very sensitive organisms. The feature of being a different color at the same temperature is meant to represent how the corals' behavioral responses differs as they are forced to constantly adapt to their changing temperature surroundings.

Q: How can I win!?

A: The game isn't set up to win. The goal of the game is to have the best quality coral possible while maintaining your company's profits. In real life, you can't exactly win because life just keeps going. The only way to end the game is to kill the corals, lose your money, or give up. It's your job to decide how you play!

Q: Why does the coral color change from red to white and then white to brown as the temperature increases?

A: Since this game is meant to simulate the real life process corals go through as ocean temperatures rise, the red to white represents the bleaching of the coral as the zooxanthellae algae gets expelled from the coral, and the white to brown represents the slow death of the coral.

Conclusion

Thank you for playing the Coral Bleaching Game!

